P600A THRU P600M

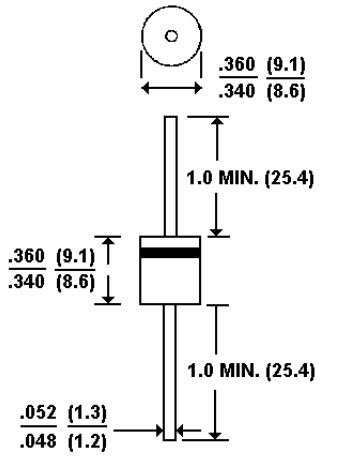
HIGH CURRENT PLASTIC SILICON RECTIFIER

VOLTAGE - 50 to 1000 Volts CURRENT - 6.0 Amperes

FEATURES

- High surge current capability
- Plastic package has Underwriters Laboratory Flammability Classification 94V-O Utilizing Flame Retardant Epoxy Molding Compound
- Void-free plastic in a P600 package
- High current operation 6.0 Amperes @ T_A=55 ¢J
- Exceeds environmental standards of MIL-S-19500/228

P600



Dimensions in inches and (millimeters)

MECHANICAL DATA

Case: Molded plastic, P600

Terminals: axial leads, solderable per MIL-STD-202, Method 208

Polarity: Color band denotes cathode

Mounting Position: Any

Weight: 0.07 ounce, 2.1 gram

MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

*@ T_A=25 ¢J unless otherwise specified. Single phase, half-wave,60 Hz, resistive or inductive load.

**All values except Maximum RMS Voltage are registered JEDEC parameters.

	P600A	P600B	P600D	P600G	P600J	P600K	P600M	UNITS
Maximum Recurrent Peak Reverse Voltage	50	100	200	400	600	800	1000	V
Maximum RMS Voltage	35	70	140	280	420	560	700	V
Maximum DC Blocking Voltage	50	100	200	400	600	800	1000	V
Maximum Average Forward Rectified Current T_A=55 ¢J	6.0							iCA A
Maximum Overload Surge Current at 1 cycle (NOTE 1)	400							A
Maximum Forward Voltage at 6.0 ADC	1.0							V
Maximum DC Reverse Current @T_A=25 ¢J	10							£g A
Rated DC Blocking Voltage @T_A=100 ¢J	1.0							mADC
Typical Junction capacitance (Note 3) CJ	150							pF
Typical Thermal Resistance (Note 2) R £KJA	20.0							¢J/W
Typical Thermal Resistance (Note 2) R £KJL	4.0							¢J/W
Operating Temperature Range	-55 to +150							¢J
Storage Temperature Range	-55 to +150							¢J

NOTES:

1. Peak forward surge current, per 8.3ms single half-sine-wave superimposed on rated load(JECED method)
2. Thermal resistance from junction to ambient and from junction to lead at 0.375"(9.5mm) lead length P.C.B. mounted with 1.1×1.1"(30×30mm) copper pads
3. Measured at 1 MHZ and applied reverse voltage of 4.0 volts

RATING AND CHARACTERISTIC CURVES

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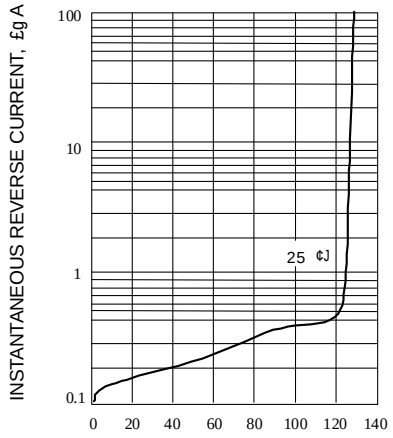

Fig. 1-TYPICAL REVERSE CHARACTERISTICS

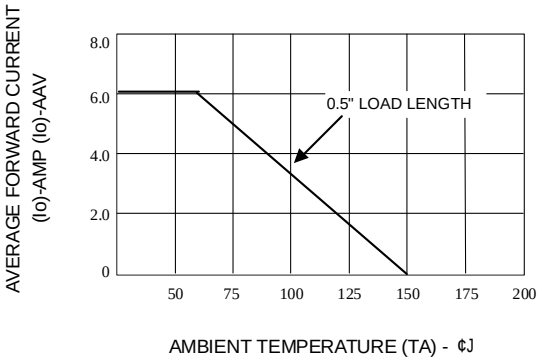

Fig. 2-FORWARD DERATING CURVE

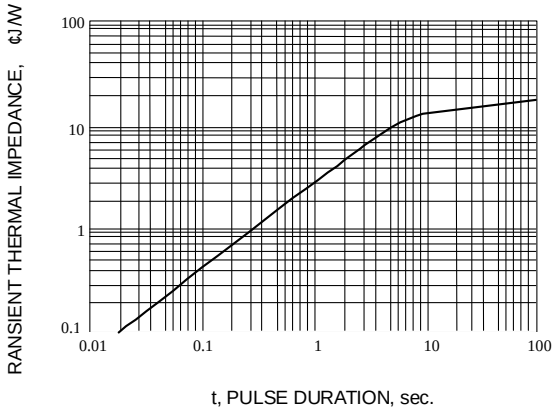

Fig. 3-TYPICAL TRANSIENT THERMAL IMPEDANCE

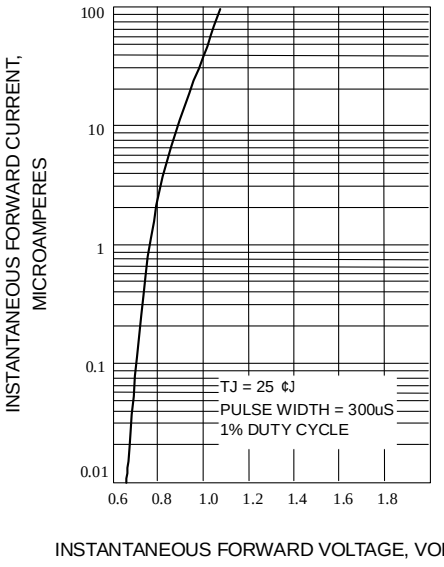

Fig. 4-TYPICAL INSTANTANEOUS FORWARD CHARACTERISTICS

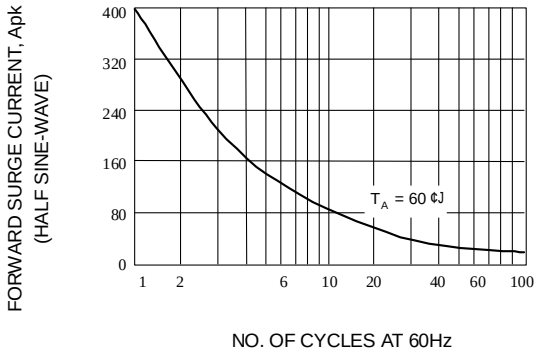

Fig. 5-MAXIMUM OVERLOAD SURGE CURRENT